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Research Article

Culturally Tailored Serious Game Design: A Revolutionary Model for Smart Agriculture

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Smart agriculture, which is culturally specific, combines education with the celebratory potlatch of our agrarian heritage. Smart agriculture, also known as precision farming, refers to the application of advanced technologies to enhance the effectiveness and efficiency of agricultural practices. The concept of serious games has gained popularity across various fields, including smart agriculture. Serious gaming is currently being considered as a new approach to address challenging farming issues and enhance farmer training, thereby stimulating technological innovation in the agricultural sector. Despite the growing interest in serious games, there is a lack of research exploring how such games can effectively communicate practical farming knowledge, particularly across different cultural practices. Research is needed to assess whether these games can effectively teach key topics, such as soil management and crop selection while respecting cultural norms. It is also crucial to evaluate whether gameplay influences players' attitudes toward sustainable farming and the extent to which they retain the knowledge in real-world contexts. This study examines a culturally specific game design model for smart agriculture, integrating educational and cultural elements to enhance learning and engagement. The game allows players to experience the farming cycle, encompassing planting, tending, and harvesting. It is intended to educate children and anyone interested in learning about the origins of their food, as well as to explain why sustainable farming is essential for future generations. The players experience farming through seasonal cycles, from decisions on seed types to soil preparation, including dealing with weather calamities. A quasiexperimental design compared two groups: a traditional learning group and an experimental serious game group. A serious game developed based on the proposed model was implemented and evaluated through pre- and posttests, statistical analyses, and heuristic evaluations to measure improvements in agricultural knowledge. The results demonstrate significant knowledge retention in the game group, particularly in precision farming, sustainability, and IoT-based solutions ($p < 0.05$). These findings highlight the potential of serious games as an innovative tool for blending traditional agricultural knowledge with modern farming practices, promoting sustainable farming education in diverse cultural settings.

Keywords: cultural context; game-based learning; serious game; smart agriculture; sustainability

1. Introduction

Culturally tailored serious games in smart agriculture are interactive approaches tailored to a specific cultural context, targeting farmers. Serious games for farmers will teach and empower them through local knowledge, practices, and challenges integrated into the gameplay, making learning more relevant and engaging. However, there are numerous pitfalls to effectively implementing these kinds of games. These include constrained access to technology among farmers, the poor state of infrastructure in some areas, financial constraints in developing games, adaptation of games to suit

different farming practices, and cultural sensitivity in diverse cultural settings [1]. The abovementioned challenges require serious planning, research, and thoughtful design to make these culturally designed games effective in encouraging sustainable agriculture globally [2].

Agriculture has revolutionized with the rise in smart agriculture as technology has played a key role in recent years. It provides the opportunity to increase food production efficiency, sustainability, and productivity. The importance of considering cultural awareness during the design and implementation of smart agricultural systems is evident as technologies worldwide continue to integrate into traditional farming methods [3]. The inclusion of cultural factors in smart agriculture design is, therefore, a consideration for sustainable practice. This cultural sensitivity in designing smart agriculture still respects and recognizes the age-old agricultural traditions and knowledge that have sustained generations in various communities [4]. Such cultural details allow technology to enhance productivity while celebrating and preserving heritage in farming communities worldwide. A significant challenge is that one-size-fits-all solutions do not work. There are specific cultural practices, environmental conditions in which people live, and farming methods that enable particular communities to function effectively with the land [5]. Adapting the technology accordingly will require knowledge of the local customs, beliefs, and practices. Cultural sensitivity in design encompasses much more than functionality, including aesthetics, usability, and community acceptance. Such a smart agriculture system would be more acceptable and sustainable if it fits with daily life and aligns with local values. The incorporation of traditional knowledge and skills by farmers, community leaders, and cultural experts into the design process further strengthens community resilience and self-reliance [6]. Including cultural considerations in smart agriculture has several advantages: This would render technological innovations more effective, aligning with local needs and preferences. Moreover, traditional knowledge is valued for preserving and revitalizing cultural heritage. This concept falls under the principle of equitable development because the approach enables any impoverished community to utilize technology without compromising their dignity or cultural identity [7–9].

A culturally tailored design thus represents a holistic approach toward smart agriculture, integrating technological innovation with cultural diversity. This bridges technology and tradition to create an inclusive, sustainable, and culturally vibrant future for global agriculture. Serious games in smart agriculture that are culturally tailored are extremely relevant for creating interactive experiences that align with the farmers' cultural backgrounds and agricultural practices. The games range from mere educational tools to more empowered, transformative platforms that reflect the local knowledge, practices, and challenges faced by farmers [10]. Immersing players in situations that reflect their agricultural contexts in the real world helps them learn and internalize better sustainable farming practices. On the other hand, several challenges are associated with using culturally adapted serious games in smart agriculture. Firstly, farmers, in general and especially those in rural areas, often have limited access to technology, including devices and internet connectivity, which hinders their ability to play these digital games [11]. What can help bridge this divide includes affordable technology solutions, the promotion of digital literacy, and the equitable distribution of resources. Other challenges include insufficient infrastructure in some regions, such as erratic electricity or internet services, which can themselves be an obstacle to the deployment of smart technologies in agriculture [12]. In the case of such regions, addressing these issues may involve partnerships with governments, NGOs, and the private sector to develop better infrastructure tailored to sustaining agricultural innovation. Financial constraints are another challenge, as the development of these culturally adapted games is a costly endeavor. Software development, content creation, localization, and support fundamentals require payment. The financial resources needed for the projects must be ensured through grants, partnerships, and innovative models if sustainability and growth are to be fostered. Farming styles vary across different cultures, and game designs must carefully adapt to local agricultural conditions [13]. What works in one region may not be suitable in another due to variations in climate, soil, and farming techniques. Adapting game mechanics and content to local realities requires thorough research, consultation with local experts, and iterative testing [14]. Cultural sensitivity also plays a crucial role in game design: games should respect and reflect local customs, beliefs, and values to foster acceptance and engagement. All this is with the proviso that cultural expression should be authentically included and stereotypes avoided by working closely with community leaders and cultural experts. The aesthetics and usability of such games should be prioritized in their design, making them visually appealing and accessible to people at all levels of digital skill. Engaging in gameplay, storytelling, and player incentives will motivate players to participate with longevity in mind [15, 16]. Involving them in the design process ensures that the games meet the needs and preferences of the local communities, enhancing their relevance and effectiveness. This participation approach also develops community members' digital literacy and technology skills. The benefits of incorporating cultural considerations into the design of a smart agriculture game (SAG) are immense. This promotes the spread of technology because it is feasible to tailor it to suit the local context and challenges [17]. It also helps preserve and promote traditional knowledge, supports equitable development, and enables communities to utilize technology while maintaining their cultural identity. Such games will drive positive changes in agricultural education and practice by overcoming various access, infrastructure, financial, and cultural sensitivity challenges. Through collaboration and innovative design, we can envision a future where smart agriculture is advanced, culturally responsive, and inclusive [18].

The culturally tailored smart agriculture systems align with the sociocultural theory in several ways: the contextual adaptation aspect of this theory emphasizes the importance of adapting to the setting where learning and behavior occur. Similarly, smart agriculture systems incorporate local cultural practices and environmental conditions, such as the

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types of crops to plant and farming techniques, to adapt to cultural norms [19]. In the sociocultural theory of community participation, learning and development are essential outcomes of social interactions. Communities in culturally appropriate systems participate in the design, implementation, and maintenance of technology to ensure that the system meets their needs and fosters local ownership. Knowledge transmission in the theory has focused on how knowledge is shared within communities. Culturally tailored systems embed traditional practices and local knowledge into modern technologies, facilitating the intergenerational transfer of valuable agricultural knowledge. In sociocultural theory, adaptive learning is adapting learning modalities in response to changes in various contexts. In this vein, culturally tailored systems utilize feedback from communities of practice to continually enhance the technology in response to shifting sociocultural environments and evolving ecological conditions [20]. Ethical considerations in sociocultural theory relate to addressing social justice issues. This means, in sum, fair distribution of technology benefits, respect for indigenous knowledge, and freedom from negative impacts on traditional practices. Overall, culturally adapted systems for smart agriculture embody the theory of sociocultural perspectives, as they uphold the importance of context, community involvement, knowledge sharing, adaptability, and ethical considerations in agricultural development.

The cultural sensitivity of smart agriculture designs extends beyond functionality to encompass aesthetics, usability, and community acceptance. In that direction, systems that work in tune with daily life and are perceived as aligned with local values are most likely to be adopted and sustained. The incorporation of farmers, community leaders, and cultural experts into the design process will ensure the integration of traditional knowledge and practices, thereby enhancing the community's resilience and self-sufficiency. Significant benefits arise from integrating cultural aspects into smart agriculture. It makes technological innovations more effective and widely accepted by tailoring them to local needs and preferences; it also supports cultural preservation, valorization, and promotion of traditional knowledge [21]. This approach will promote equitable development, enabling marginalized communities to utilize technology while preserving their cultural identity and practices. There are, however, several challenges to developing culturally adapted serious games for smart agriculture. The major hurdle is the limited access to technology, particularly in rural and remote areas [22]. Most farmers lack the necessary devices or internet access to engage in digital games; therefore, strategies are needed to bridge this digital divide through access to affordable technologies and enhanced digital literacy. When the infrastructure is generally inadequate, deployment becomes even more complicated. These could be addressed through partnerships between governments, NGOs, and the private sector to develop better infrastructure that will pave the way for agricultural innovation.

The financial aspects are yet another limiting factor, as the development of culturally specific games has become so costly regarding investment in software, content creation, and ongoing support [23]. Securing a variety of funding sources, including grants and partnerships, is crucial to sustaining projects like these. This also means, however, that game designs need to be adapted to each local agricultural context, and the only way these can be relevant and effective is by conducting extensive research and consultation with experts locally [24]. This raises several questions about the culture of the people; therefore, sensitivity in designing the games will be required, considering local customs and values, to ensure cultural acceptance within the community. This can be achieved through close collaboration between community leaders and cultural experts to prevent stereotyping and ensure cultural accuracy. Aesthetics and usability: Games must be visually appealing and not overly challenging to navigate, as participants may have varying digital skills. It encourages activities that are rewarding and call for longer-term involvement. Local communities should be consulted during the game design phase. This significantly improves the game's ability to address needs, preferences, relevance, effectiveness, and applications, ultimately enhancing skills in digital literacy and technology at the local level. Integrating cultural concerns within the games of smart agriculture enhances their effectiveness by getting the technology in tune with the local context, preserves cultural heritage, and allows equitable development [25]. Addressing access, infrastructure, financial, and culturally sensitive challenges will accelerate positive changes in agricultural education and practice. We can only create a truly advanced and culturally responsive future of smart agriculture by marrying technology with tradition, ensuring that farmers and their communities worldwide have a sustainable and prosperous future.

These systems can, therefore, support culturally sensitive, sustainable agriculture at both national and international levels by simply embedding these principles. Keeping that in mind, the following questions were aligned in this study:

1. How will serious game design for smart agriculture consider cultural factors and preferences?
2. How can one determine the effectiveness of culturally tailored serious games in promoting sustainable farming practices?

The paper has been organized as follows. Section 2 presents a comprehensive literature review, drawing on previous studies related to the impact of culture in smart agriculture and serious games for smart agriculture. While Section 3 presents the proposed model for designing a culturally tailored game, Section 4 gives details regarding design, low-fidelity, and high-fidelity prototypes. Furthermore, the experimental setup details in Section 5 present a discussion of the results, and the article concludes in Section 6.

2. Literature Review

This section presents the relevant literature on cultural context, factors for smart agriculture, serious games related to smart agriculture, design models, and the game developed by the researcher, along with its validation.

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2.1. Cultural Context With Its Impact and Factors for Smart Agriculture. Societies that highly value technological progress, such as most Western countries and many of the rapidly growing economies in Asia, would adapt more quickly to new technologies in agriculture [26]. These are cultures in which technology is often considered one of the primary drivers of progress and improvement. In other words, if there is a place like Silicon Valley for information and communication technology, the same place could emerge as a hub for ag-tech innovations [27]. With startups innovating in this space, the frontiers of what is possible with smart agriculture continue to expand, leading to significant improvements in farm technology. On the other hand, in those few cultures where traditionalism is deeply ingrained, resistance to change and the adoption of new technologies remain relatively strong. This largely stems from a profound belief in the dependability and efficiency of long-established methods that have been refined over generations. Farmers in rural parts of India or Sub-Saharan Africa might consider smart agriculture suspicious [28]. This could be a threat to traditional farming knowledge or, worse still, a disregard for the social structures that are deeply ingrained in these ancient practices. The potential resistance to change in this regard remains one of the mainstays to establishing new agricultural technologies in the more conservative regions of the world. In the economically well-endowed parts of the world, like Western Europe, North America, and increasingly parts of East Asia, the infrastructural and financial wherewithal for smart agriculture is more accessible [29–31]. These areas typically have established supply chains, internet access, and support systems that facilitate the seamless integration of advanced technologies. For instance, the Netherlands has established itself as one of the leading countries in smart agriculture. This is partly due to its robust infrastructure, substantial investment in agricultural research, and a supportive culture that encourages innovation in farm technology. In contrast, there are quite a few obstacles to the adoption of smart agriculture in developing regions [32].

Additionally, low capital availability can impose financial constraints on the adoption of capital-intensive technologies, such as GPS systems, automated machinery, and data analytics tools. Added to this is the complication brought about by poor infrastructure and spotty internet access in large swaths of rural Africa and Southeast Asia, where smart agriculture options are further limited. Overcoming such hurdles will likely require a combination of international aid initiatives, governmental subsidies, and partnerships with technology providers to make many of these technologies more accessible and affordable. More educated and competent regions have a greater competitive advantage in reaping benefits from smart agriculture. For instance, countries like Germany and Japan, with rigorous educational systems supported by technical training, provide the largest pools of professionals who can develop, manage, and optimize these technologies. These countries are also home to research institutions and universities that are strong drivers of agricultural technology innovations [33]. Less-educated regions may be significantly hindered in adopting and utilizing smart agriculture technologies effectively. Without training or support systems, farmers in these regions are unable to comprehend or practically apply these complex tools and systems. That is where outreach and education programs play a key role. These NGOs and development agencies provide the much-needed training and inputs to fill these educational gaps. Technology adoption tends to be more communal in cultures that are more collectivist-oriented, where ties within the community and family are strong. For instance, in parts of Latin America and Asia, agricultural cooperatives or shared farming may facilitate the sharing of smart technologies. These cultures often weigh the collective benefit above individual gain, which can influence how technologies are adopted and integrated. Conversely, regions that firmly hold traditional values may give higher prominence to established agricultural practices. This is typified by rural Japan or even parts of the Middle East, where traditional farming practices are deeply rooted in cultural heritage. The challenge here is introducing new technologies that complement these traditions rather than harming them, making the transition smooth and acceptable. The introduction of precision agriculture tools, for example, can be made to align with traditional practices and values, helping to smooth the adoption process [34].

Government support is another crucial factor for the successful integration of technology in agriculture. Favorable policies, finance, and infrastructure create an enabling environment that fosters the adoption of smart agriculture [34]. The EU Common Agricultural Policy enshrines provisions to encourage technological advancements and sustainability in agriculture. Similarly, countries such as the Netherlands and China have implemented policy settings that provide favorable conditions for agricultural innovation by offering subsidies and research support, which may be necessary to stimulate growth in smart agriculture. On the other hand, regions with strict regulations or inadequate support may be hindered from adopting these technologies. Inefficient bureaucracy, lack of incentives, or obsolete policies suppress the process of innovation and retard the pace at which these technologies are adopted and the establishment of challenging new practices. Social dynamics also play a significant role in the adoption of smart agriculture technologies. Community leaders' peer influence and endorsements significantly impact the decision-making process. Influential members of any community will promote the use of new technologies, and others are more likely to follow their example. Conversely, when the community is skeptical or resistant, the adoption process will reverse. At the international level, particularly within developing regions, NGOs and various advocacy groups often play a crucial role in promoting smart agriculture through education, technical support, and financial assistance, helping to overcome adoption barriers. Organizations such as the Bill & Melinda Gates Foundation and the World Bank have already begun using technology to improve agriculture with involvement from underserved regions. To be truly helpful, smart agriculture technologies must be adapted to local climate and environmental conditions [35]. For instance, precision irrigation systems must be adapted to regional water supply and soil moisture

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conditions. In this respect, smart agriculture can only succeed if it understands and integrates local environmental factors. Those cultures that are firmly focused on ecological sustainability are most capable of adopting practices that balance the ecology. For example, Scandinavian countries strongly espouse concepts of sustainability. As a result, technologies that reduce environmental impact and promote green practices have been introduced. The adoption and diffusion of smart agriculture are closely tied to their respective cultural contexts. The prevailing cultural, economic, and social diversities may shape smart agriculture to better meet local needs and conditions. When cultural attitudes, financial constraints, educational levels, and other contextual factors are taken into consideration, stakeholders can facilitate the better applicability and increased dissemination of smart agriculture [36, 37]. This approach will enhance agricultural productivity while promoting the dissemination of benefits from technological development across various farming communities worldwide.

2.2. Serious Game and Smart Agriculture. Serious games are interactive digital games designed for purposes beyond pure entertainment, including education, training, and problem-solving [38]. In recent years, the concept of serious games has gained popularity across various fields, including smart agriculture. Smart agriculture is also referred to as precision farming, which means applying advanced technology to make farming more effective and proficient [8, 9, 30, 39–41]. Serious gaming is currently being considered as a new approach to address challenging farming issues and enhance farmer training, thereby stimulating technological innovation in the agricultural sector. These games are designed for specific purposes beyond mere entertainment; they incorporate game-like elements into more immersive learning and problem-solving experiences. Key elements of serious games include interactivity, modeling real situations, some educational content interwoven through gameplay, and feedback mechanisms giving the player immediate feedback from their actions that can further their learning and decision-making [42]. Serious games often utilize simulations of actual agricultural systems, including crop management and the operation of farm equipment. They could also involve role-playing, where players assume different agricultural roles to solve specific problems or participate in decision-making scenarios that require informed choices based on available data. Serious games can be of immense use in smart agriculture. They can also do some simulated farming for training and education, wherein farmers can try a practice or technique in a virtual space without the risk of failing in the field [43, 44]. A game can simulate irrigation schedules and fertilizer applications, as well as various methods of pest control, in a safe environment that allows farmers to test different methods in a valuable way before applying them in the field. Serious games will also enable users to afford technical training in using sophisticated farm machinery, such as drones, automated tractors, and precision planters, making them proficient in these technologies more rapidly. Serious games can also act as decision-support tools by simulating various farming scenarios, from variable weather to market conditions and even pest outbreaks [44]. These games enable users to experiment with the outcomes of different crop yield and farm profit strategies, thereby informing their decision-making processes more effectively. They will also contribute to the interpretation of data; for example, smart agriculture technologies will provide farmers with vast amounts of complex data, such as from satellite images or sensors, that need to be interpreted to make informed decisions on irrigation or pest control.

Serious games may finally foster innovation and research by crowdsourcing solutions, involving large numbers of players in finding solutions to agricultural challenges such as soil degradation or climate adaptation. Players' collective ideas may also help in research and development. These games also serve as testing grounds for new agricultural technologies that developers can then refine before implementing them in the real world [45]. Community engagement and awareness are other benefits. Serious games on sustainable farming practices would also inform players of their pros and cons. For example, some games can be applied in real life to indicate to players the benefits of organic farming or conservation tillage. Such games serve as educational tools, helping to engage young people and the general public in enjoyable and easily understandable agricultural activities. This educates the next generation of farmers and agricultural scientists by teaching concepts in crop rotation, soil health, and water management. Serious games offer several advantages in smart agriculture applications by making learning and engagement more interactive, practical, and easy to comprehend for users with complex agricultural concepts. This opens up possibilities for innovation within the game by allowing for the testing of different strategies and technologies with minimal risk. These games also offer a cost-effective method of training, eliminating the need for expensive physical simulations and on-site training. Additionally, serious games enable farmers to make informed decisions based on the modeling of agricultural scenarios or data analysis [46]. However, there are some challenges in integrating serious games into smart agriculture. For example, game design that successfully models complex agricultural systems requires skilled game developers with knowledge of agrarian systems. Ensuring realism and usefulness in simulations may become problematic. The games need to be made accessible and easy to use for all farmers, regardless of their varying technical skills. The integration of serious games with existing smart agriculture technologies and data systems is a complex process, as the potentials must be harmonious with other utilities and platforms. Updates in the game that sustain user motivation and, accordingly, engagement are also critical; therefore, step-by-step improvements to the new game are necessary to keep it interesting and relevant. Serious games can help improve smart agriculture–based training, decision-making, and community engagement. Interrelating through interaction and game-based learning provides valuable insights for innovation and the adoption of new technologies in agriculture. With continuous improvements in both technology and gaming, serious games have great potential to make substantial contributions to smart agriculture in the future, benefiting farmers, researchers, and the broader agricultural community [47].

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Table 1 summarizes SAGs aiming at different aspects of agriculture in various cultures. The design and development of such games require frameworks that provide guiding principles for building a comprehensive product. Several generic frameworks and models have been proposed for educational, mobile, and SAGs, specifically for mobile applications. The closest model is available for English vocabulary in the Arab context. However, there are some limitations in this model from the following perspectives: (a) specific requirement gathering by engaging the relevant stakeholders, (b) missing usability perspective, (c) no structured approach for gamification, and (d) the proposed model has not been validated.

3. Serious Game Design Model for Smart Agriculture

A serious game development framework encompasses all the above elements, ranging from various game types to flexibility in response to change and the ability to assess, learn, and improve the learning process. Frameworks and methodologies for serious game design help developers cooperate on one coherent solution. Researchers have proposed several serious game design models for smart agriculture; however, serious attention is required for a serious game design model in a smart agricultural setting within a cultural context [30, 48]. Therefore, this research proposes a serious game design model for smart agriculture in a cultural context by combining elements such as agrielements, educational objectives, cultural context, technology and innovations, usability (including usability guidelines and usability testing), and sustainability and impact, as shown in Figure 1 by considering the model by [30] as a base. The domain experts reviewed and validated the model using IRR. IRR is defined as the degree of agreement between raters or judges. If there is complete agreement, the IRR = 1 (or 100%); if there is no agreement, the IRR = 0 (0%). Various experts have stated that a consent percentage of 75%–90% is an appropriate level of agreement [30].

3.1. Components for Serious Game Design Model. This section presents the game's SAG design using the serious game design model. The SAG is aimed at supporting learning smart agriculture in the cultural context of Pakistan [49]. This prototype has been developed based on the proposed theoretical model and extensively evaluated by engaging a considerable sample of public sector primary school students. The design of SAG strictly involves the components of the proposed theoretical model, including agrielements, educational objectives, cultural context, technology and innovations, usability (as guided by usability guidelines and testing), and sustainability and impact. SAG has been developed from the planning to the testing stage as a software project. First, the requirements were gathered from stakeholders through a preliminary survey and a literature review to design the interfaces and modules, considering the elements of the theoretical model.

3.1.1. Agrielements

3.1.1.1. Crop Management. Designing a serious game for crop management would involve taking the gamer through various experiential situations and teaching them the importance of agriculture and resource management. There may be several types of crops that have different needs and contributions [49]. Seasons and climates change, too, which may eventually affect planting. Players must manage resources, including water, fertilizers, energy, and economic factors such as shifting market prices and investment opportunities [50]. Problem incorporation—the challenges involved, such as pest management and soil health—is further complemented with educational tutorials and case studies to facilitate a more profound understanding among the players. A user-friendly interface, detailed feedback, and multiplayer collaboration options enhance engagement and learning, making the game a comprehensive tool for investigating sustainable agriculture practices.

3.1.1.2. Farm Machinery and Tools. Farm machinery and implements are essential in today's farming, as efficiency and productivity have increased significantly. The vital machinery is the tractor, which provides versatile power for various farm operations, such as plowing, planting, and harvesting. Combine harvesters simplify the collection of crops from the field; at the same time, the plow and harrow prepare the soil for planting by breaking and smoothing the earth. Planters and seeders plant the seeds fast and precisely, while irrigation systems ensure the exact amount of soil moisture needed [51, 52]. Specialized equipment, such as tillers, cultivators, and balers, makes cultivating the soil, controlling weeds, and harvesting crops more efficient. If these tools are implemented effectively, they could contribute increased yield, reduced labor cost, and improved handling of resources, thereby becoming vital in any successful and sustainable farming venture [53].

3.1.1.3. Weather and Seasons. This involves integrating weather and seasons into a serious game design model, adding dynamism and learning to enhance gameplay and learning outcomes. That means, in a game, the player himself has to make decisions on the types of weather and the changes in seasons and their consequences related to crop management [54]. For example, this can be achieved by simulating seasonal cycles, requiring the player to strategize planting and harvesting around periods of optimal time, such as spring for cool-season crops and summer for warm-season crops. Weather events, such as rain, drought, or frost, would be a great way to introduce realistic effects on the health and yield of crops, prompting players to adjust irrigation methods and protection measures or even change their strategies wholesale. These features will enable gamers to learn how, in real life, weather and seasonal variability have affected agriculture. This will develop their critical thinking and strategic planning skills. This creates a far more enthralling and engaging experience, simulating the actual, rigorous conditions typically experienced by farmers, thanks to the ability to incorporate simulated weather patterns and seasonal changes.

3.1.1.4. Irrigation. The irrigation systems would, therefore, form one of the pivotal game mechanics in any significant game design model. This was because irrigation systems

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TABLE 1: Summary of games and applications for smart agriculture.

Game	Description	Limitations	Year	Approach	Link
Farming Simulator	Manage and operate a virtual farm where they can grow crops, raise livestock, and handle various farming tasks	Complexity, graphics, and performance, limited scope	2020	Sandbox approach, realistic simulation	https://www.farming-simulator.com/ Giants Software's official site
Plantix	Plant disease diagnosis and management	Image quality	2015	Image recognition technology	Plantix official website
Harvest Moon	Managing a farm, growing crops, raising livestock, and interacting with the local community	Repetitiveness, graphics, and aging	2020	A mix of simulation elements, role-playing, and open-ended gameplay	Harvest Moon official website
Agriprener	Manage a virtual farm or agricultural enterprise, making decisions	Complexity, accessibility, realism vs. playability	2020	Simulation-based approach	IFPRI website
FarmVille	Social farming simulation game	Microtransactions, time-dependent mechanics, repetitiveness	2020	Casual, social gaming approach	FarmVille official page
Stardew Valley	Farming simulation role-playing game	Graphic style, complexity, time management	2018	Open-ended sandbox approach	Official Stardew Valley website
Farmcraft	Time management and a simulation game	Repetitiveness, graphics and design, time management, pressure	2020	Time management approach combined with a farming simulation	Green Sauce Games' official website
Harvest Harmony	A farming simulation game that blends traditional farming mechanics	Niche appeal, complexity, graphics, and content	2021	Farming simulation with community-building elements	Starwind Games official website
AgriQuet	A farming simulation game that combines traditional agricultural management	Development and updates, niche appeal, graphics, and performance	2020	A hybrid approach combining farming simulation	
My Time at Portia	Manage a workshop and engage in farming activities	Performance and complexity	2018	Combines farming simulation, RPG, and crafting elements. Players build relationships with NPCs and explore a rich, open world	Time at Portia Official Site https://www.mytimeatportia.com/
Garden Paws	Features an open-world setting with a focus on customization and community interaction	Graphics and content	2018	Uses a blend of farming simulation and life simulation, with a focus on player customization and community interaction	https://www.gardenpaws.game/
Littlewood	RPG and farming simulation game where players rebuild their town after a catastrophic event	Graphics and gameplay depth	2020	Combining town-building and farming simulation with RPG elements	https://www.littlewoods.com/
Staxel	A voxel-based farming simulation game that blends building and farming elements	Graphics and performance	2018	Uses a voxel-based approach combining farming simulation with exploration and building elements	https://playstaxel.com/
Farm Together	A cooperative farming game where players work together to manage and expand a farm	Repetitiveness and multiplayer focus	2018	Farming simulation with real-time progression and casual gameplay	https://farmtogether.com/
Harvest Moon: Light of Hope	Recent entry in the beloved *Harvest Moon* series	Graphics and gameplay depth	2021	It focuses on rebuilding and managing a farm within a charming, story-driven world	https://www.hmnation.com/

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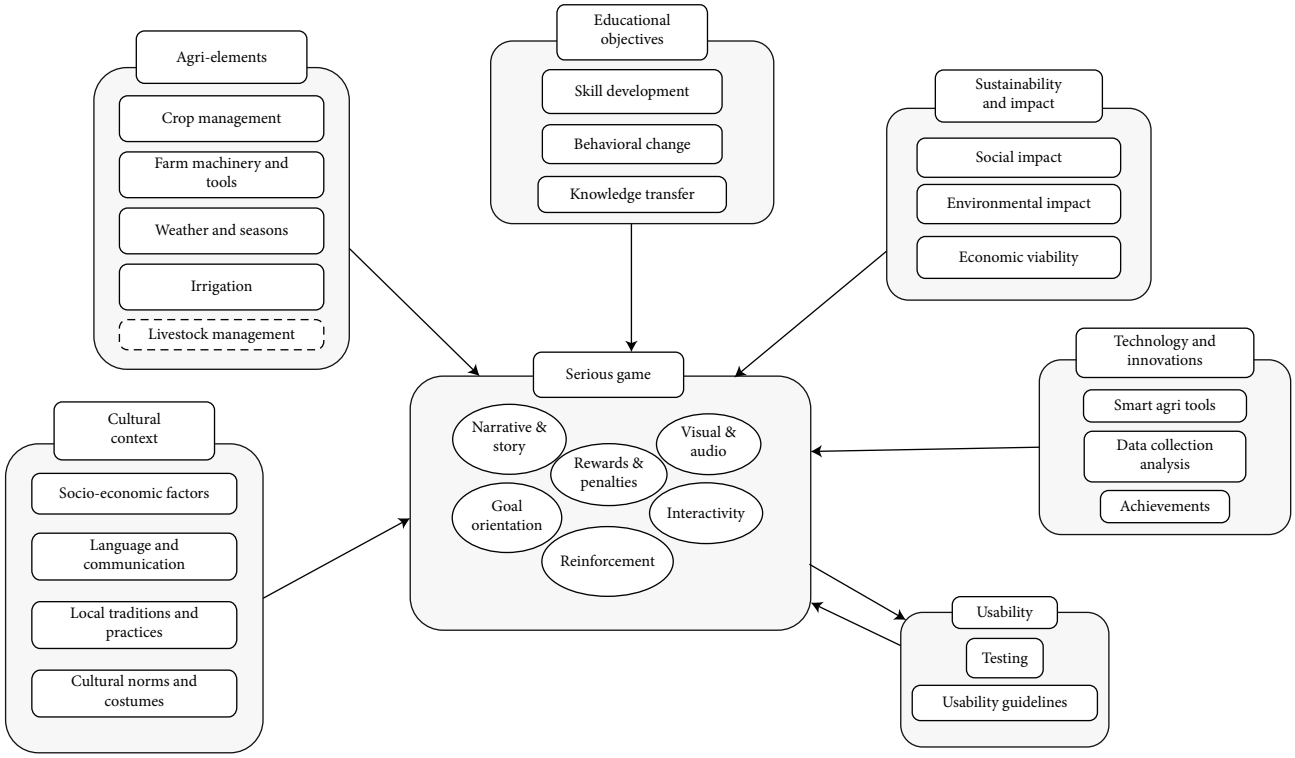


Figure 1: Serious game design model for smart agriculture in a cultural context.

underpinned the importance of prudent water use tied to agricultural activity. Players were to select and manage various irrigation methodologies, ranging from classical sprinklers to drip systems and advanced automated solutions. Each of these had particular advantages and disadvantages. The game can simulate various water availability conditions, including droughts and excessive rainfall. This would imply adjusting irrigation strategies to maintain the crops at their optimal health and yield maximum yields. The players have to balance the use of water with environmental sustainability. They need to understand the practices related to water conservation and recognize how their choices affect crop productivity and ecological systems. In that respect, integrated irrigation mechanics teach the player more effective ways to manage water, think strategically, and solve problems, much like in real-world agriculture [54].

3.1.2. Educational Objectives

3.1.2.1. Skill Development. In serious game design, skill development is a primary educational goal, encompassing practical knowledge and competencies through interactive gameplay. The game was individually designed to develop strategic planning, resource management, and problem-solving skills. Independent players would engage in activities related to real-life challenges, such as maintaining crop cycles, optimizing resource use, and adapting to unplanned events, including pests or sudden changes in weather conditions [55]. As players advance through the levels, they develop critical thinking skills by analyzing the outcomes of their decisions and refining better strategies based on feedback and key performance metrics. The game can also integrate educational modules and tutorials to enhance their understanding of specific concepts, such as soil health, irrigation techniques, or the principles of market economics. This game thus fills the gap between theoretical knowledge and practical application, interacting with and efficiently bridging the divide between theory and practice by simulating real-life situations and providing opportunities for practice and reflection. Valuable skills apply to players in agriculture and beyond.

3.1.2.2. Behavioral Change. A key educational objective in serious game design models is to create behavioral change in players' attitudes and actions regarding real-world practices. The game may provide you with a simulated environment that lets you know what happens due to your decisions and hence takes into consideration your adoption of more sustainable and efficient behavior [56]. For example, it could show players the consequences of resource overexploitation and poor environmental conservation, which may stimulate their curiosity and drive them to adopt better practices. In each iterative play, gamers will experience how a positive strategy, such as water conservation, integrated pest management, or soil conservation, contributes to the overall benefit. This integration of feedback mechanisms, rewarding players for positive actions in sustainability and punishing them for highly detrimental practices, further reinforces positive behavioral change. Ultimately, this immersive and interactive nature of the game can lead to a profound understanding of the significant ways responsible behavior truly matters, motivating players to apply these principles in their lives and current agricultural practices.

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3.1.2.3. Knowledge Transfer. Knowledge transfer would be one of the primary educational objectives of any serious game design model, wherein the player should be able to apply the concepts and skills learned within the game to the real world [57]. The game is designed to replicate realistic scenarios and problems, allowing players to rehearse and internalize key concepts in crop management, resource allocation, and sustainability practices. This game translates theoretical knowledge into practice through detailed tutorials, interactive challenges, and real-world case studies. For instance, players can apply concepts learned from irrigation best practices in the game to actual agricultural practices or other resource allocation scenarios. Additionally, the reflective elements of the game, such as debriefing and knowledge checks, will enable players to consolidate their learned knowledge and understand how decisions made in the game relate to real-world applications. This approach has increased the relevance and applicability of the content, enhancing the assurance of transfer from the virtual environment to real, tangible situations [57].

3.1.3. Cultural Context

3.1.3.1. Socioeconomic Factors. A serious game design model that integrates socioeconomic factors into the cultural context can provide a multilayered learning experience and be more realistic. With its diverse economic scenarios and cultural practices, this game can reflect the various conditions and challenges faced by different communities worldwide. For example, players would be able to manage farms in regions with varying economic conditions, progressing from the most affluent and technologically advanced to developing areas with limited resources. It should model socioeconomic aspects, such as access to markets, availability of labor, and access to finance, that would impact decisions made by players regarding resource management, crop selection, and investment strategy. The cultural context can also be significant, where the game represents a locality's farming traditions and customs that influence agricultural methodology. This may range from traditional knowledge regarding crop rotation and irrigation systems to culturally relevant crops. By doing this, the game provides a richer and more immersive experience, fostering a deeper understanding and respect for diverse agricultural practices and socioeconomic conditions. Eventually, this will help players appreciate the diversity of challenges and solutions to farming in the real world, which are embedded in socioeconomic factors and cultural contexts, equipping them to make informed decisions on various issues within their community or others [30, 57].

3.1.3.2. Language and Communication. In serious game design, embedding language and communication within their cultural contexts is an essential component of an effective and engaging experience in smart agriculture. The game should support multiple languages in view of linguistic diversity [42]. Contents, instructions, and dialogues are to be translated with their cultural adaptation. This clearly states that a mere translation of the text is needed, along with considering cultural references to tune in with the players in their native languages. Localized agriculture-related practices, challenges, and traditions should also be represented in the game to vary scenarios and make them more relevant and interesting. Features for communication can also be provided to accommodate different literacy levels or learning preferences through the use of visual aids, symbols, and voice commands. The social features can include forums where players can discuss and share knowledge and strategies, as well as collaborative missions. Finally, embedding feedback mechanisms that are sensitive to cultural concerns and refreshing the game regularly based on user contributions may help keep it fresh and relevant. By addressing these facets, the game should be much better positioned to engage users and facilitate the uptake of smart agricultural practices across diverse cultural backgrounds [30, 42].

3.1.3.3. Local Traditions and Practices. In designing serious games for smart agriculture, the incorporation of local traditions and practices is quite indispensable in creating a culturally relevant and practical experience [58]. The gaming should reflect farming practices specific to target regions by including local crops, farming techniques, and seasonal cycles. This involves situations relevant to actual farming and the challenges faced by players, such as rice cultivation in states where rice is a staple food. It also takes into account cultural beliefs about agriculture, such as Thanksgiving and other tribal agricultural coming-of-age rituals. Address regional agrarian challenges, such as scarce water or soil fertility, for instance, and ensure the game provides workable solutions for local conditions [59]. The involvement of local experts and people in the design process would enhance the authenticity and relevance of the game. Additionally, facilities for knowledge and experience sharing at the regional level, such as forums or group missions, foster a sense of community and provide a user-friendly platform for disseminating valuable information. By embedding these traditions and practices within the game framework, the developers create a significantly more effective educational tool that resonates with players and promotes the adoption of smart agriculture practices [30].

3.1.3.4. Cultural Norms and Costumes. These are essential aspects to consider in serious game design for smart agriculture. Cultural norms and traditional costumes must be incorporated into the design of such an engaging yet culturally sensitive experience. This is achieved by combining local customs, social practices, and values into the gamified scenarios, ensuring the content is relevant to the target audience and aligns with their daily agricultural practices [60]. It also draws character designs and inspiration for game environments from traditional clothing, adding a touch of realism to make the game more interesting and further allowing players to see their culture represented in it [59, 61]. This is important and requires careful consideration, as it necessitates treating the subject with sensitivity and respect. This means conducting thorough research or consulting cultural experts to ensure an accurate representation of a culture. It is also educational in promoting culture and

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preservation by adhering to cultural norms and traditional costumes, fostering greater engagement among players as they feel a sense of familiarity and pride. This makes the game both relevant and respectful, enhancing learning while instilling an appreciation for the diverse farming practices and traditions [30, 59, 61].

3.1.4. Technology and Innovations

3.1.4.1. Smart Agritools. In the design of a serious game dealing with smart agriculture, integration with smart agritools and technologies is essential; this will make the game impactful and educative [62]. Precision agriculture devices, drones, and soil sensors are among the technologies that should be integrated into the game, as recent advances and innovations have significantly enhanced this field. Such an incorporation of technologies will enable the game to let the real-world applications allow players to know how smart tools enhance agriculture interactively [63]. Players would be able to do things like flying a drone virtually to monitor crop health, applying precision irrigation techniques using simulated soil sensors, or analyzing data for informed decisions regarding planting and harvesting [64]. This would help players learn not only about the practical uses and benefits of those technologies but also prepare them for real-life applications. Moreover, the advanced innovations included in such a game may stir up interest and curiosity that eventually could encourage players to become familiar with how these tools can be put into use to solve contemporary agricultural problems [65]. Therefore, it can help bridge the gap between theoretical knowledge and its implementation in real life through practical exposure to the use of smart agritools [66]. Hence, it will serve as a platform for learning and innovation in smart agriculture.

3.1.4.2. Data Collection Analysis. In serious game design for smart agriculture, the integration of features in data collection and analysis is significant in providing realism and education [67]. The incorporation of these advanced technological elements in the game can simulate processes related to the collection and interpretation of agricultural data, including soil moisture, crop growth rates, and weather patterns. Such activities might include virtually collecting data from sensors and then analyzing the same through interactive dashboards to make data-driven decisions for improving farming practices [68]. This helps players understand how real-world data collection and analysis tools enhance agricultural efficiency and productivity. Players might adjust irrigation schedules, predict pest infestations, and improve crop yields with that imaginary data. This game provides education on the role and significance of data in modern agriculture, offering practical experience in making decisions based on the data provided. By integrating data collection and analysis into gameplay, the gap between theory and practice may become narrower, and this game can serve as a valuable tool for learning about smart agriculture technologies and innovations.

3.1.4.3. Achievements. Incorporating achievements into serious games designed for smart agriculture adds a layer of seriousness to motivation and reward, enabling technology innovations to engage players more effectively and promote learning. Achievements could be designed to reflect key milestones and accomplishments related to the practice of smart agriculture, such as the successful implementation of precision irrigation systems, achieving optimal crop yields through data analysis, or utilizing cutting-edge agricultural implements like drones and soil sensors. These can be tracked through in-game mechanisms that help players gain badges, certificates, or unlockable content, providing them with tangible recognition of progress and expertise. Furthermore, this methodological approach to the game will motivate exploratory and skill development in various dimensions of smart agriculture, and it will also reinforce learning by applying technological solutions to specific achievements. This pedagogic experience can be extended and made more engaging by incorporating a formal achievement system into the game, enabling players to gain insight into how technological innovations contribute to modern agricultural practices [49].

3.1.5. Usability

3.1.5.1. Testing. In serious game design for smart agriculture, usability testing ensures the game is intuitive, accessible, and effective for the target audience. It involves assessing the ease with which users can navigate the game, understand its mechanics, and achieve learning objectives related to smart agriculture. Players play the game in front of testers, while usability tests gather feedback on the design of interfaces, user controls, and overall gameplay experiences. Such feedback allows the identification of various challenges or frustrations that users might encounter, and so developers make necessary adjustments to improve functionality and the user experience of the game [30, 69]. This might include a usability test indicating that certain agricultural tools or data analysis features are not intuitive; through simplification or clearer instructions, these elements might be revised. The benefits of early detection of usability issues in the design process enable the game to be optimized, allowing complex agricultural concepts and technologies to be effectively communicated in an engaging and user-friendly manner. This iterative test and refinement process has finally provided the educational value of the game with a boost, serving as a more effective tool for learning and innovation in smart agriculture [30].

3.1.5.2. Usability Guidelines. This could also serve as an essential guide for designing a serious game for smart agriculture, ensuring easy usability, engagement, and effectiveness in addressing complex agricultural concepts that may be involved. These usability guidelines include simplicity, clarity, and consistency of the gameplay. The game interface should be unburdening and straightforward, with intuitively obvious operations that minimize users' cognitive load, enabling them to easily access and utilize the various features offered through the interface, such as precision farming tools or data analytics dashboards. Secondly, clear instructions and effective feedback mechanisms are equally important, whereby players will immediately and understandably react

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to the actions taken by others, helping them understand how their decisions have implications for agricultural outcomes. Using consistent design elements and terminology within the game can create a cohesive experience that avoids confusion, allowing users to feel more confident. Additionally, variable test sizes, color-blind modes, and customizable controls enhance accessibility features, making this game inclusive for all users. Following such usability guidelines will enable serious game developers to create an instructive game that is an entertaining and enjoyable game for smart agriculture participants [30].

3.1.6. Sustainability and Impact

3.1.6.1. Social Impact. Design a serious game for a smart agricultural community considering the impact of social issues. This game is aimed at raising awareness of sustainable farming practices and demonstrating how smart agriculture technologies can address pressing environmental and social issues, including resource conservation, climate change, and food security. The game will teach players how to distribute water more effectively, reduce chemical runoff, and manage soil health by incorporating scenarios that demonstrate the economic benefits of these practices. It can also enhance community involvement by simulating collaborative farming activities and demonstrating how technology increases the community's resiliency and economic well-being [70]. Other elements of this can include features that promote social responsibility and ethical decision-making, encouraging players to consider the broader ramifications of their farming practices. This, in turn, highlights the social benefits of smart agriculture, fostering effective change and enhanced contributions towards sustainable and more equitable agriculture.

3.1.6.2. Environmental Impact. Other main themes to be considered while creating a serious game for smart agriculture include ecological impact, which will contribute to increased environmental sustainability by demonstrating the effects of modern agricultural technologies [71]. It should also include scenarios illustrating how smart agriculture practices promote environmental conservation, such as optimizing water application, reducing chemical use, or enhancing soil health. Technologies such as precision irrigation systems, drone monitoring of crops, or soil sensors are implemented to minimize environmental footprints and promote sustainable resource management. It would ask the player to perform basic irrigation management tasks to prevent water waste and conduct data analysis to reduce pesticide use, demonstrating the potential for methods to achieve an environmental advantage. The game can also help players understand the long-term effects on biodiversity and ecosystem health that result from sustainable farming, allowing them to appreciate the importance of incorporating environmental considerations into agricultural decision-making. The game showcases the potential of smart agriculture to benefit the environment with high efficacy, thereby potentially motivating players to adopt and contribute to creating a more sustainable and resilient agricultural system [30].

3.1.6.3. Economic Viability. Integrating economic viability into smart agricultural serious game design will demonstrate how modern agriculture can be economically viable and have a positive impact. Real-world economic scenarios should be simulated in the game to demonstrate the cost-effectiveness and resulting financial benefits of implementing smart agriculture technologies [72]. This would include budgeting features that allow players to gauge the return on investments concerning various technologies and analyze the economic consequences of decisions relating to farm management. A player could, for instance, utilize in-game tools to identify cost savings from precision irrigation systems or enhance profitability through optimized crop management. The game educates users on how smart agriculture can enhance economic resilience and profitability by incorporating scenarios that demonstrate the short-term financial benefits of sustainable practices. Scenarios that provide insights into marketplace trends, pricing strategies, and risk management would further explain the financial benefits of adopting advanced agricultural technologies. Ultimately, it could focus on the economic viability of smart agriculture, enabling players to adopt innovations that also lead to financial success in the agricultural sector [30].

3.1.7. Serious Game/Gameplay Mechanics

3.1.7.1. Serious Game Design Framework. The serious game design framework serves as the central component of the proposed model, functioning as an interactive tool for delivering culturally contextualized agricultural education to school-aged children. The game was developed to facilitate skill acquisition, knowledge transfer, and behavioral improvement through immersive gameplay that aligns with local culture, language, and farming practices [40].

3.1.7.2. Farming Concepts and Content Structure. The game is structured into multiple progressive levels, each designed to introduce a specific farming concept grounded in real-life agricultural practices. The core topics covered include crop management, such as crop cycles, sowing, and harvesting; water usage and irrigation practices for efficient resource management; the use and purpose of various farm machinery and tools; and weather-related decisions to enhance understanding of timing and environmental impact. Each level is enriched with interactive tasks, visual storytelling, and culturally familiar scenarios that help children contextualize agricultural knowledge in a way that is meaningful and memorable [30, 40].

3.1.7.3. Game Mechanics and Cultural Adaptation. The game incorporates several core mechanics to enhance learning and motivation. Cultural relevance was prioritized during development, ensuring the integration of local language, customs, attire, and environmental elements to foster a stronger emotional connection with learners. Narrative-driven storytelling was used across levels to guide children through realistic rural farming situations. Each level presents goal-oriented tasks such as selecting the appropriate tool or adjusting irrigation systems. These structured goals are designed to mirror real-life challenges and decision-

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making processes in the agricultural sector. A scoring system, along with badges and stars, was implemented to reward successful performance and reinforce correct behaviors. Visual and audio elements tailored to local settings further support learning by utilizing familiar symbols, terms, and sounds. The gameplay encourages interactivity through decision-making, quizzes, and hands-on tasks, with reinforcement occurring through immediate feedback and gradual task repetition to promote retention [30].

3.1.7.4. Learning Assessment: In-Game and Surveys. To measure the game's educational effectiveness, both in-game and external assessment strategies were employed. Internally, the game tracks learner progress through scores, task completion, level advancement, and embedded quizzes at the end of each level. Externally, a pre- and postsurvey evaluation was conducted with participating children. The presurvey assessed prior knowledge of agriculture, while the postsurvey measured knowledge gained after completing the game. This before-and-after comparison provided quantifiable insights into the game's impact on knowledge acquisition and concept understanding [40].

3.1.7.5. Integration With the Model. The serious game component aligns closely with all other dimensions of the proposed model. Agricultural knowledge is embedded directly through game tasks. Cultural elements are reflected in the visuals, language, and setting of the game. Educational goals are met through structured engagement and formative assessments. Technology is leveraged to deliver interactive experiences and performance tracking. Usability is ensured through an intuitive and child-friendly interface. The design also supports sustainability goals by integrating lessons on eco-friendly practices and responsible resource usage. In summary, the serious game acts as a culturally adaptive and educationally effective learning platform that transforms traditional farming knowledge into an engaging digital experience. Through level-based tasks, culturally rooted storytelling, rewards, and a combination of in-game and survey-based assessments, the game enables children to gain practical agricultural knowledge while enjoying the learning process. The design successfully combines cultural relevance with measurable educational outcomes, offering a valuable model for interactive rural education [40].

3.1.7.6. Selection of Cultural Elements. The selection of cultural elements for the game was based on both observation and community understanding. The development team began by reviewing standard farming practices, rural school settings, and locally recognizable visuals. Field visits were also conducted in nearby villages to observe how agriculture is practiced and how children interact with their surroundings. The goal was to choose visual and content elements that children would instantly recognize. This was done to make the game feel familiar and reduce the difficulty of understanding new concepts. Alongside cultural familiarity, the educational value of each element was also taken into consideration. Only those aspects that could support the learning of agricultural or environmental topics in a simple and age-appropriate way were included [30].

3.1.7.7. Input From the Local Community. Community involvement played an important role during the design process. Local teachers, parents, and farmers were informally consulted to guide the content and tone of the game. Teachers helped identify what farming topics children were less familiar with and gave feedback on whether the content was suitable for the age group. Parents and farmers contributed by sharing their everyday experiences, tools they use, farming methods, and seasonal challenges. Although the process was not part of a formal participatory design study, these discussions helped ensure that the game was grounded in real rural life rather than just theoretical knowledge. Their input improved the cultural accuracy and practical value of the farming activities shown in the game [30].

3.1.7.8. Testing Cultural Relevance. To determine if the game felt culturally appropriate and understandable to the children, testing was conducted in both classroom and community learning settings. Developers observed how children reacted during gameplay, paying particular attention to their attention, facial expressions, and ease in completing tasks. Children were also asked simple follow-up questions, such as whether the game reminded them of real life or if they understood the tools and words used in the game. These reflections, along with the children's natural comments and behavior, helped measure the game's cultural impact. Many children pointed out things they had seen at home or in the fields, such as traditional clothing or the way crops were grown. This confirmed that the game matched their lived experiences and was culturally relatable [30].

3.1.8. Expert Validation and Model Assessment. To ensure the credibility of the proposed game design framework, a panel of six experts was consulted. Their feedback was assessed through an interrater agreement (IRA) approach using a simple percentage calculation. Table 2 shows that the model components received a high level of consensus from all experts. Four out of five categories achieved a perfect agreement score of 100%, while the *technology and innovations* component received a slightly lower score, with five experts in agreement and one offering an alternative opinion, resulting in an 83% consensus. The overall agreement stood at 90%, reflecting strong consistency in expert evaluations. Experts provided several constructive recommendations for refining the game model. These included incorporating audio pronunciation support throughout the learning stages and implementing a system that requires learners to revisit and repeat lessons after making multiple mistakes. Additionally, the experts stressed the importance of conducting high-fidelity prototype evaluations with real users (teachers and students) before deployment to uncover and address any usability or instructional design concerns at an early stage.

4. SAG Based on the Incremental Model

This section outlines the design and development of the SAG using an incremental model approach. The objective of SAG is to promote awareness and learning of smart

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TABLE 2: Expert agreement analysis.

No.	Components	Expert A	Expert B	Expert C	Expert D	Expert E	Expert F	Level of agreement
1	Educational objectives	1	1	1	1	1	1	6/6 (100%)
2	Cultural context	1	1	1	1	1	1	6/6 (100%)
3	Sustainability and impact	1	1	1	1	1	1	6/6 (100%)
4	Technology and innovations	1	1	1	1	0	1	5/6 (83%)
5	Usability	1	1	1	1	1	1	6/6 (100%)
Total								0.90 (90%)

agriculture practices among school students, contextualized within the cultural environment of Pakistan. The game was conceptualized based on a theoretical framework (illustrated in Figure 1), emphasizing educational usability, sociocultural relevance, and interaction design.

4.1. Game Development Process. Development followed a software engineering lifecycle, beginning with requirements gathering through surveys and literature reviews. These insights shaped the design of the game's interface and modules. The Unity game engine was used to develop the prototype for Android devices. The game's core modules were structured around key agricultural themes: farming techniques, environmental sustainability, modern agricultural tools, resource optimization, and agricultural market dynamics. Each module integrates culturally familiar visuals, local terminology, and region-specific content to ensure relatability. Gameplay features realistic scenarios, problem-solving challenges, and decisions that align with sustainable agricultural practices. To test the usability and engagement of the product, two types of prototypes were developed: low fidelity and high fidelity, and evaluated iteratively before final user testing.

4.1.1. Low-Fidelity Prototype Design. The low-fidelity prototype focused on structural flow and user navigation. It was created to demonstrate the user interface layout and the intended interactions without implementing final graphics or animations. Upon launching the game, users encounter a splash screen featuring a "Tap to Play" option, followed by a main menu offering access to several features: *Farmer Profiles*, *Achievements*, *Garage*, *Credits*, and a prominent *Play* button.

Once the user selects *Play*, a mode selection screen appears, allowing them to either begin a new session or continue from a previously saved game. Gameplay modes are designed to unlock progressively as the user advances. After choosing a mode, the player is directed to a level selection interface. Initial access is restricted to Level 1, with later levels (e.g., Level 2 and Level 3) becoming available based on performance.

During gameplay, users navigate through multiple screens. Upon successful completion of a level, a "Level Completed" screen provides the option to move forward. In case of failure, the user is presented with a retry option. A pause menu allows for resuming, restarting, or exiting to the home screen. A settings panel enables adjustment of music and sound effects, while an exit confirmation screen prevents accidental closures. Figures 2a, 2b, 2c, 2d, 2e, 3a, 3b, 3c, 3d, and 3e illustrate key elements of the prototype flow.

4.1.2. High-Fidelity Prototype Design. The high-fidelity prototype represents the finalized game design with full visual elements, animations, and audio effects. It was developed to simulate the complete user experience, ensuring visual appeal and intuitive interaction. When users launch the game, an animated loading screen transitions into a rich main menu offering multiple choices, including *Select Farmer*, *Achievements*, *Garage Access*, and *Play Game*.

Users interact with unlockable game modes and a visually organized level selection interface, which reflects their progress. Upon selecting a level, players are immersed in gameplay that mirrors real-world agricultural environments. Gameplay includes decision-making scenarios related to farming practices, resource usage, and environmental impacts in Figures 4a, 4b, 4c, 4d, 4e, 5a, 5b, 5c, 5d, and 5e.

On level completion, users receive performance feedback. Success leads to progression, while failure offers opportunities for retry. A pause interface provides control over game flow, and a customizable settings menu allows changes to sound and music levels. A confirmation screen ensures intentional exits. These features collectively enhance the educational impact, engagement level, and technical polish of the game [30, 40].

4.2. Usability Testing and Heuristic Evaluation of SAG. To examine and enhance the user interface of the SAG, a combination of heuristic evaluation and the think-aloud technique was employed. These methods helped uncover interface limitations that could hinder effective learning for users in the smart agriculture domain. By applying these techniques, the development team gained valuable feedback to improve the interface design, while usability specialists were able to identify obstacles that might affect user engagement or comprehension. Usability evaluation was conducted in two phases: first, during the low-fidelity prototype phase and, later, with the high-fidelity version of the game. A custom-designed heuristic checklist, tailored specifically for educational games within culturally sensitive agricultural contexts, was developed and used for the evaluations, as no prior framework existed for this particular setting. Based on standard guidelines, the study determined that between five and eight evaluators would provide reliable feedback. For the low-fidelity assessment, evaluators included teachers and a game developer, whereas teachers and students reviewed the high-fidelity version. Additionally, teachers took part in the think-aloud sessions to express their real-time thoughts while interacting with the high-fidelity prototype.

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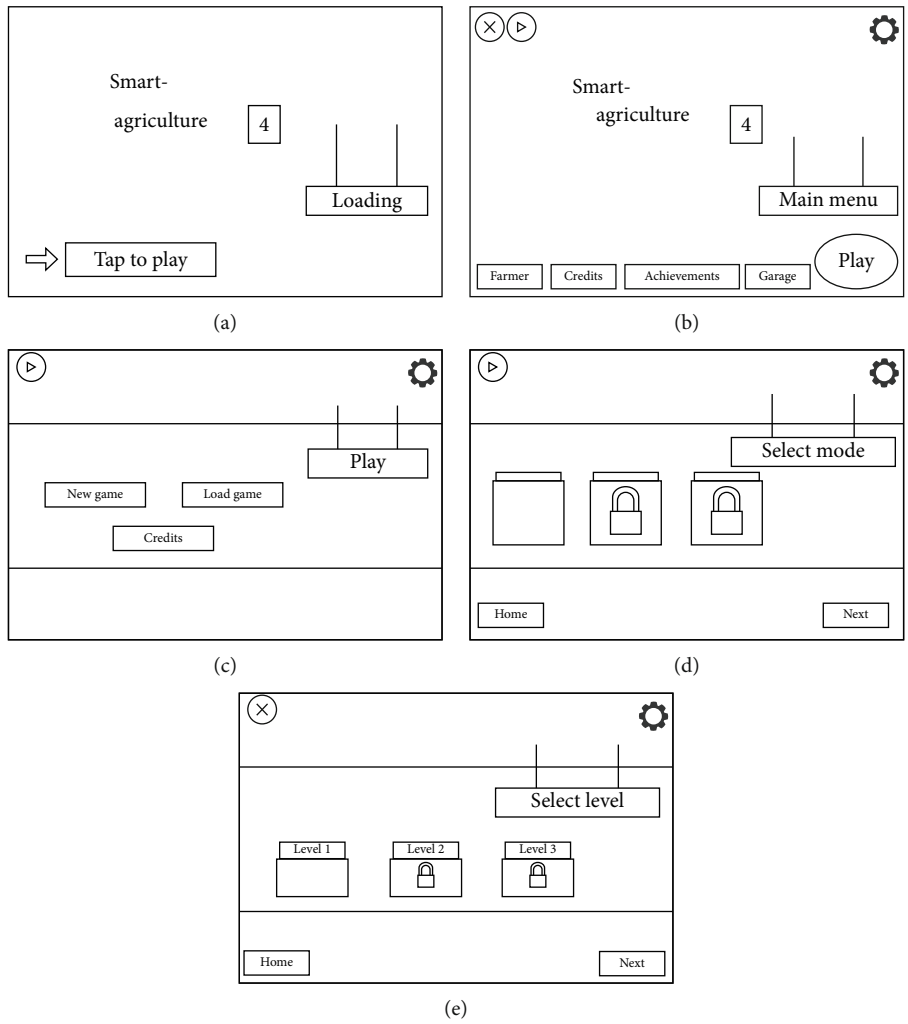


FIGURE 2: Low-fidelity prototype of SAG. (a) Low-fidelity Prototype 1. (b) Low-fidelity 2. (c) Low-fidelity 3. (d) Low-fidelity 4. (e) Low-fidelity 5.

The rationale behind selecting teachers was their familiarity with evaluating educational content, while the game developer offered insights into technical and gameplay aspects. The low-fidelity prototype evaluations were conducted in the school principal's office using printed designs and mock-ups. The high-fidelity prototype evaluations took place both in classrooms (with students) and again in the principal's office (with teachers).

Following the heuristic and think-aloud evaluations, feedback was collected through direct notes and audio recordings. Participants' verbal insights were recorded, and the researcher later transcribed the conversations from Urdu to English to facilitate qualitative analysis. To enhance the accuracy and credibility of this process, a triangulation strategy was adopted—cross-verifying transcriptions between the primary researcher and a coresearcher. This ensured reliability by comparing interpretations drawn from multiple data sources and methods. The conclusions were synthesized by integrating user observations, questionnaire responses, and verbal feedback collected during the sessions.

4.2.1. Ethical Considerations. This study involved the participation of school-aged children and was conducted in accordance with established ethical standards. Ethical clearance for the study was obtained from the Institutional Review Board (IRB) of the relevant institution. The research design, informed consent process, and data handling procedures adhered to national and institutional ethical guidelines. Before data collection commenced, informed consent was obtained from the school administration, the participants' parents or legal guardians, and the students themselves. Participation in the study was entirely voluntary, and the anonymity and confidentiality of all participants were maintained throughout the research process.

4.2.2. Sample Randomization and Bias Control. The sample consisted of 40 students from grades five and six, aged between 12 and 15 years, selected through convenience sampling from a school in Lahore, Punjab. To ensure a fair comparison and minimize selection bias, students were randomly assigned to two groups. One group ($n = 20$) served as the experimental group and engaged with the serious

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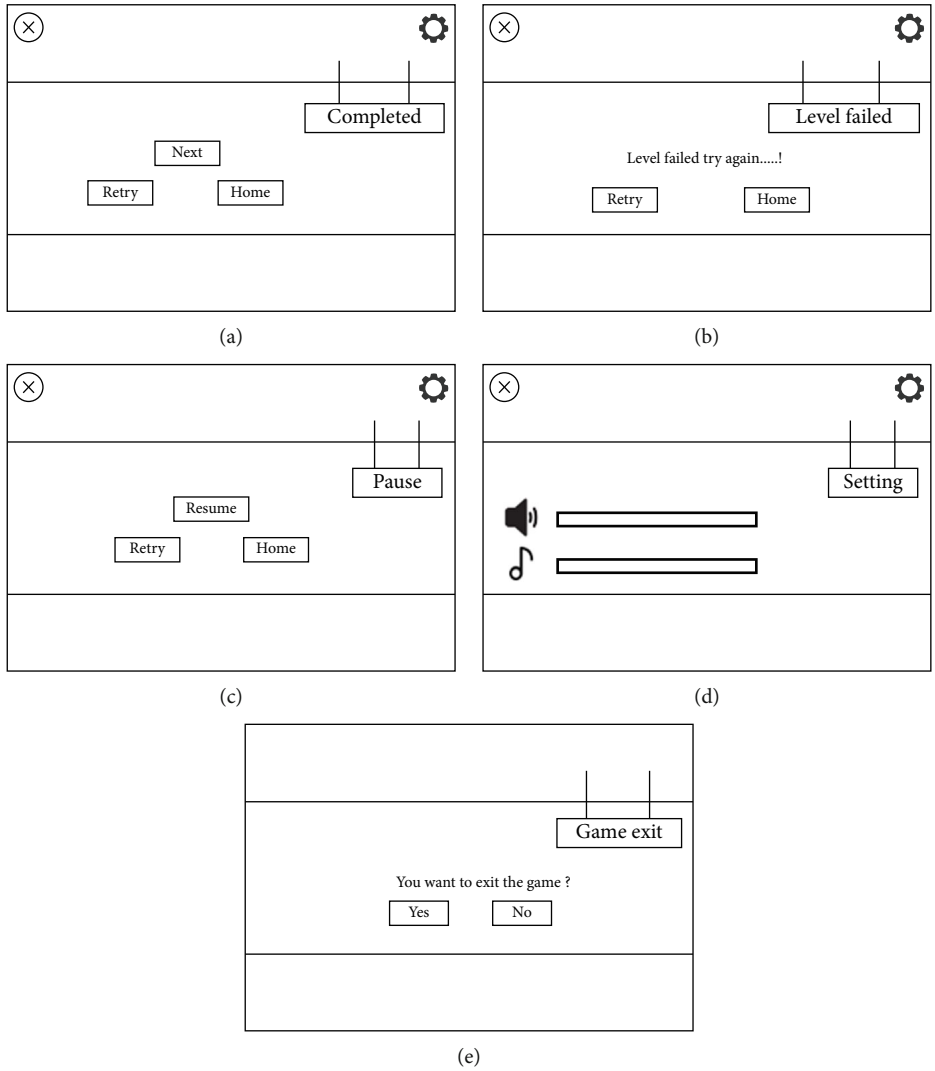


Figure 3: (a) Low-fidelity 1. (b) Low-fidelity 2. (c) Low-fidelity 3. (d) Low-fidelity 4. (e) Low-fidelity 5.

game, while the other group ($n = 20$) acted as the control group and received traditional classroom instruction. Random assignment was conducted using a simple random number generator, with no researcher interference.

To mitigate potential bias and uphold the integrity of the experimental conditions, both groups were instructed in separate classrooms under the supervision of independent facilitators. These facilitators followed standardized instructional protocols to ensure that the interventions were implemented consistently across sessions. The physical and procedural separation of the groups helped preserve the internal validity of the research and minimized the influence of external factors on learning outcomes.

4.2.3. Statistical Results and Effect Sizes. A pretest/posttest quasiexperimental design was employed to evaluate the effectiveness of the SASGDM-based serious game in improving student learning outcomes. An independent samples t -test was used to compare the posttest mean scores between the experimental and control groups. The experimental group recorded a posttest mean of 33.8 with a standard deviation (SD) of 4.12, whereas the control group had a posttest mean of 26.1 with a SD of 4.80. The results of the t-test were $t(38) = 5.44$, with a p value of less than 0.001. The 95% confidence interval for the mean difference was calculated to be between 4.89 and 10.11. The effect size, measured using Cohen's d, was 1.72, indicating a significant practical impact of the serious game intervention.

These statistical findings suggest that students who participated in the serious game experienced a significantly greater improvement in agricultural knowledge compared to those in the control group who received traditional instruction.

4.2.4. Test Domains and Instrument Validation. The assessment instruments for both the pretest and the posttest were developed in collaboration with agricultural education experts and reviewed by the school's senior headmaster to ensure content validity. Each test included four open-ended questions, with each question carrying 10 marks. The questions were designed to assess core areas defined within the SASGDM framework, including crop

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FIGURE 4: High-fidelity prototype of SAG. (a) High-fidelity 1. (b) High-fidelity 2. (c) High-fidelity 3. (d) High-fidelity 4. (e) High-fidelity 5.

management and seasonal planning, irrigation methods and water usage efficiency, identification and use of farm tools, and weather forecasting as it relates to agriculture. An example of a pretest question was "list three tools used in farming and explain the purpose of each." The posttest included more application-focused questions, such as "given a drought scenario, how would you manage water use on a small farm?" The design of these questions was aimed at evaluating not only factual recall but also critical thinking and problem-solving skills. Both instruments were subjected to expert review by three agricultural specialists and were piloted with a group of five students to evaluate clarity, appropriateness for the age group, and instrument reliability. Feedback from this pilot was used to finalize the assessment tools before their full deployment in the study.

4.2.5. Cultural Integration in Game Design. The serious game developed for this study was carefully designed to reflect the cultural and agricultural realities of rural Pakistan. Several contextual elements were incorporated to ensure that the learning environment within the game felt familiar and authentic to the student participants. Local crops, such as wheat, cotton, and sugarcane, were integrated as core components in farming simulations that involved planting and harvesting. Character designs included traditional clothing styles to reflect regional norms, and the in-game environment was modeled after typical rural village settings, featuring elements such as mud houses, irrigation canals, hand pumps, and cattle sheds. Additionally, cultural festivals, including Basant and local harvest celebrations, were embedded within the game's narrative as interactive storytelling elements. The language used for in-game instructions

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FIGURE 5: (a) High-fidelity 1. (b) High-fidelity 2. (c) High-fidelity 3. (d) High-fidelity 4. (e) High-fidelity 5.

and dialogue was adapted to the local vernacular to ensure accessibility for the target student population.

To validate the cultural relevance of the game, informal consultations were conducted with local teachers, parents, and farmers during the design phase. During usability testing, students expressed that many aspects of the game resembled their surroundings and practices. Statements such as "this reminds me of my home" and "we use these tools on our farm" were frequently noted, confirming that the game successfully captured the cultural context and was perceived as relatable by its intended users. To evaluate the usability and learner experience of the serious game, a structured experimental study was conducted involving 40 schoolchildren from Grades 5 and 6, aged between 12 and 15 years, in a school located in Lahore, Punjab, Pakistan. Out of these, 20 students were part of the experimental group that interacted with the serious game prototype developed using the SASGDM framework. In comparison, the remaining 20 students were assigned to a control group that received traditional instruction. The experimental setup was designed using a quasiexperimental research approach and implemented over a period of 10 weeks. Each group was facilitated in separate environments to avoid external influence, and independent supervisors ensured integrity and consistency throughout the experiment.

To collect usability feedback, multiple methods were employed. These included direct observation of the students during gameplay, informal interviews with participants, and structured posttest Likert-scale surveys that captured students' perceptions regarding engagement, satisfaction, ease of use, and perceived learning outcomes. The surveys were carefully developed with input from agricultural education experts and aligned with the learning goals of the game. Additionally, participants were asked subjective questions that assessed their ability to apply concepts covered in the game.

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From the usability data collected, several key findings emerged. Students generally appreciated the interactive nature of the game, particularly the visuals, cultural references, and hands-on activities. Many children expressed enjoyment in completing game-based tasks such as planting, harvesting, and using simulated tools. However, some students initially struggled with control navigation and level progression, especially in earlier versions of the prototype. This feedback highlighted the importance of intuitive game flow and age-appropriate design. Based on these observations, several changes were implemented to improve the game. These included simplifying control mechanics for more straightforward navigation, enhancing the visual clarity of in-game icons and instructions, and adjusting the level sequence to ensure smoother transitions between tasks. Furthermore, additional audio cues and help prompts were integrated to guide players through unfamiliar activities. These refinements were designed to improve accessibility, reduce confusion, and enhance overall engagement among younger learners.

In summary, the usability testing phase provided critical insights that led to the refinement of the game's interface, instructional design, and user experience. The combination of observational data, student feedback, and structured posttest evaluation confirmed that the game was both engaging and educationally effective, while also identifying areas for iterative improvement. These insights contributed to ensuring that the game aligned with the SASGDM framework's emphasis on usability, interactivity, and culturally grounded learning experiences.

4.3. Prototype Assessment Results: Low Versus High Fidelity. This section discusses the outcomes of the experts' heuristic assessment and the think-aloud approach conducted in the context of culturally tailored smart agriculture. The results of the heuristic evaluation were presented in summary or tabular form, highlighting usability concerns related to the low- and high-fidelity prototypes of the SAG. Descriptive analysis was performed using Microsoft Excel, with experts responding with "yes," "no," or "not sure" to evaluate the prototypes. For the low-fidelity prototype, most assessment items achieved an average score of 100%, while a few items scored 80%. Overall, the heuristic evaluation and think-aloud approach revealed that the low-fidelity prototype integrated most of the required cultural context, agricultural education, and language adaptation features, making it user-friendly.

Similarly, the high-fidelity prototype was validated by five experts and 20 participants, including farmers and students, who used the SAG on tablets and smartphones. Responses were recorded using the same "yes," "no," or "not sure" scale on a survey questionnaire adopted from [30, 40]. The "not sure" response indicated uncertainty about specific features. While most heuristic components scored an average of 100%, one item received an average score of 80% from experts, and five items received an average score of 90% with "not sure" responses from participants. Additionally, pre- and postsurvey results demonstrated significant improvements in participants' knowledge and engagement levels. The postsurvey results indicated that the system was perceived as highly relevant, easy to navigate, and effective in providing culturally tailored agricultural education. Overall, the heuristic evaluation, think-aloud approach, and survey results confirmed that the high-fidelity prototype successfully incorporated key cultural, educational, and user-centered design elements.

5. Study Framework and Evaluation Results

5.1. Experimental Setup. This subsection outlines the experimental framework used in the study, detailing the design strategy, target participants, and steps taken during the evaluation process.

Quasiexperimental methodology was adopted for this quantitative investigation. The study was structured by dividing participants into two distinct cohorts: one group followed a conventional instructional method (referred to as the control group), while the other engaged with the SAG (referred to as the experimental group). Both groups completed assessments prior to and following the intervention in the form of pretest and posttest evaluations. This approach allowed for a comparative analysis of the learning outcomes achieved through traditional instruction versus game-based learning.

The experiment was conducted in three stages: (1) For the pretest, the subject expert prepared a question paper consisting of four questions (each worth 10 marks) and gave it to all groups for 30 min. (2) The treatment was administered only to the experimental serious game group for 10 weeks, while the other group received their routine treatment. (3) A posttest was conducted on all groups for 30 min, using a separate question paper to obtain results after the treatment. Figure 6 illustrates the detailed experimental procedure performed in this study.

5.1.1. Instrument. Two question papers (one for the pretest and one for the posttest) were designed by the school's senior subject experts to evaluate students separately in the pretest and posttest. The posttest question paper was slightly more challenging than the pretest. All students from the two groups were required to complete the same question paper to evaluate their performance level, covering significant topics such as agrielements, educational objectives, cultural context, technology and innovations, usability, sustainability, and their impact.

5.1.2. Population Sample. This study selected 30 students randomly from the third grade of the school. The average age of the students was 11 years. There were two groups of 30 students, and the test was administered in separate rooms. Three teachers separately supervised the class while the principal administered the examination. The researcher was present to observe the whole process.

5.1.3. Pretest. The data were gathered at the start of the fall 2024 term for the pretest, whereas the exam was administered to all groups in extracurricular classes. After completing the exam, the concerned teachers collected the answer sheets and prepared the results. Ultimately, the result sheets were countersigned by the principal and then handed over to the researcher.

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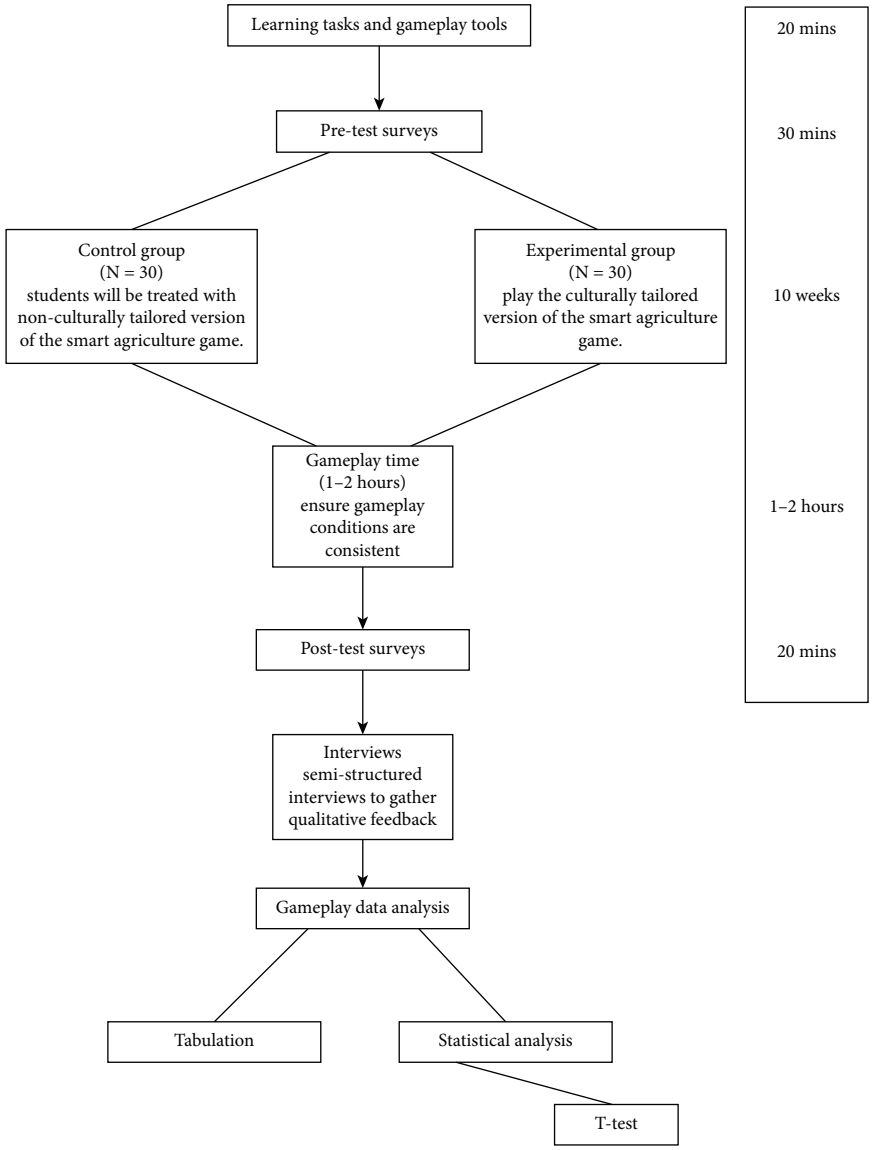


Figure 6: Experimental procedure.

5.1.4. Treatment. This section describes the treatment procedure administered to all groups participating in the experiment. All the groups followed the treatment separately through their respective methods, and the groups had to cover a similar syllabus for 10 weeks. The principal supervised the entire process, which was conducted with all the groups. The researcher guided the teachers and students on their issues during the learning and assessment process [30]. The group-wise treatment description is mentioned below:

1. Control group: This group did not receive the experimental treatment but was taught the conventional method without the use of technology.

2. Experimental serious game group: This group received the treatment of SAG developed by the researcher. Before starting the treatment, students received a brief overview of the designed game for language learning and were instructed on how to use it effectively to achieve optimal learning outcomes. Students enjoyed performing on the game's new features, that is, color scheme, interactive interface, sound effects, and pictorial presentation of the content.

5.1.5. Posttest. Both groups of 30 students were seated according to their assigned plan in different rooms for the posttest. Students had to solve another question paper because the experimental serious game group was only treated with SAG, whereas the other groups were treated with the traditional methods. The question paper was slightly more challenging than the pretest, but its nature was similar to that of the pretest. The researcher observed all activities and guided teachers and students as needed. The posttest data were gathered in the spring term of

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2025, and the exam was conducted during extracurricular classes. The teachers collected the answer sheets and prepared the results. Ultimately, the principal countersigned the result sheets and handed them over to the researcher.

To assess students' baseline understanding and perceptions related to smart agriculture and culturally relevant digital learning, a structured pretest was administered prior to the intervention. The pretest was administered through a digital form. It consisted of 14 closed-ended questions designed to evaluate familiarity, confidence, attitudes, and readiness across five thematic domains: educational objectives, cultural context, sustainability and impact, technology and innovations, and usability.

Each question used a single-response Likert scale, offering ordinal options such as the following:

- Very knowledgeable–very unknowledgeable
- Very familiar–very unfamiliar
- Strongly agree–strongly disagree
- Very likely–very unlikely

Each response was assigned a numerical value from 1 to 5, with 5 representing the most positive or informed response (e.g., "*very familiar*" or "*very confident*") and 1 representing the most negative (e.g., "*very unfamiliar*" or "*not confident at all*").

Each domain corresponds to a set of related questions, including educational objectives, which cover baseline knowledge of smart agriculture practices and the ability to apply technology (e.g., IoT and precision farming). The cultural context included questions on familiarity with local farming methods, indigenous tools, and regional relevance. Sustainability and impact included responses on environmental understanding, community benefit, and social attitudes. Technology and innovations assessed confidence in using digital tools and familiarity with smart farming concepts. Usability captured responses on game engagement, expected ease of use, and preference compared to traditional methods.

Each participant's responses were grouped by domain. For example, if three questions were used to assess technology and innovations, the average score for those three responses was calculated for each student. Then, all individual student averages for that domain were aggregated and averaged across the group ($n = 20$ for each group) to produce a single mean value per domain.

Thus, a score of 4.6 for usability in the experimental group means that, on average, students in that group selected responses close to "very engaged," "much better than traditional methods," and "very likely to recommend" when asked about their expectations and preferences for using a game-based learning tool. The same procedure was applied to calculate domain scores for the control group.

These computed averages were reported to provide a baseline comparison of the traditional group and the experimental group before the learning intervention began. This method ensured consistency in scoring, statistical validity, and alignment with the SASGDM framework domains.

TABLE 3: Pretest results of all the groups.

Agrielements	Traditional group	Experimental serious game group
Educational objectives	3.8	4.4
Cultural context	3.9	4.2
Sustainability and impact	3.6	4.5
Technology and innovations	3.6	4.3
Usability	3.8	4.6

5.2. Results and Interpretation. This section outlines the performance comparison between the two participant groups—the control group and the experimental group that used the serious game, using both tabular data (average scores) and statistical insights.

5.2.1. Summary Table

5.2.1.1. Pretest Results. Table 3 displays the average scores obtained by both groups across four targeted learning dimensions: *cultural understanding*, *sustainability awareness*, *technological insight*, and *system usability*, which are illustrated in Figure 7. The data reveal that the group consistently engaged with the serious game and achieved higher scores in all categories, indicating a stronger initial grasp of the educational objectives. Notably, the traditional group recorded its highest pretest score in *cultural understanding*, with an average of 4.1, as illustrated in Figure 7.

5.2.1.2. Posttest Results. Table 4 presents the average results of the traditional group and experimental serious game group for educational objectives, cultural context, sustainability and impact, technology and innovations, and usability, which are illustrated in Figure 8. The pretest results showed no significant difference in performance between the two groups. However, in the posttest, the experimental serious game group demonstrated a substantial improvement across all areas compared to the control group. Figure 8 provides a graphical representation of the posttest results, highlighting the enhanced performance in the experimental serious game group. A t-test was conducted to analyze the differences between the pretest and posttest scores of each group. The results revealed a statistically significant improvement in the experimental serious game group, indicating the effectiveness of the serious game in enhancing learning outcomes compared to traditional methods.

5.3. Descriptive Analysis of Pretest and Posttest Scores for Both Groups. Below is a summary of the mean, SD, and score improvement for each group.

5.3.1. Control Group. The traditional group demonstrated an overall improvement in posttest scores, with a mean increase of 0.22, as shown in Table 5. The highest improvement was in sustainability and impact (+0.3). The SD is slight, suggesting low variability in responses.

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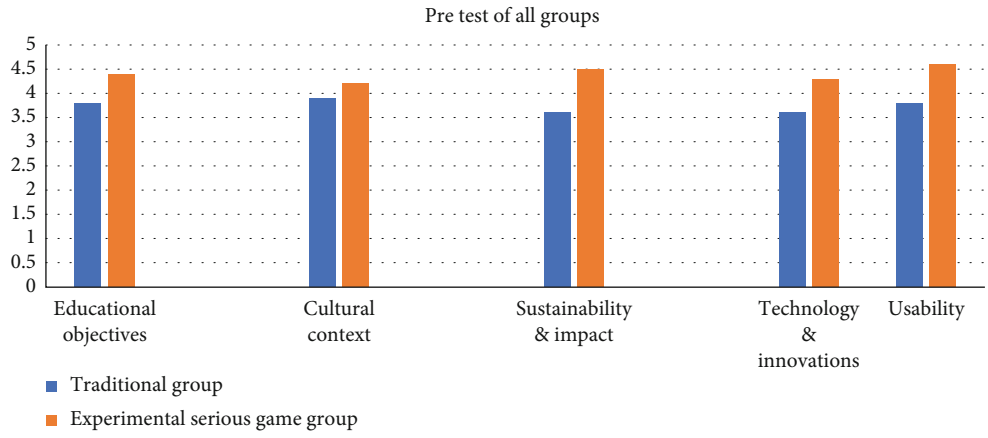


Figure 7: Bar chart of all the pretest groups.

TABLE 4: Posttest results of all the groups.

Agrielements	Traditional group	Experimental serious game group
Educational objectives	4.0	5.6
Cultural context	4.1	5.4
Sustainability and impact	3.9	5.6
Technology and innovations	3.8	5.5
Usability	4.0	5.7

5.3.2. Experimental Serious Game Group. The experimental serious game group also demonstrated improvement, with a mean increase of 0.16, as shown in Table 6. The highest improvements were in educational objectives, cultural context, and technology and innovations, each by +0.2. The SD is slightly lower than that of the traditional group, indicating greater response consistency.

5.3.3. Comparison of Both Groups. Both groups demonstrated statistically significant improvements ($p < 0.05$), indicating that the learning interventions were effective. The control group had a higher mean improvement (+0.22) than the experimental group (+0.16), but both groups demonstrated learning gains. The experimental group had higher initial scores, indicating that participants were more familiar with the subject matter before the intervention. The low p values (< 0.05) confirm that these changes were not due to random chance, as shown in Table 7.

Both control and experimental serious game methods effectively improved knowledge. The control group showed slightly higher improvements, possibly due to lower pretest scores. The experimental serious game group had a higher starting knowledge, meaning they benefited somewhat less but still saw gains. Given the statistical significance in both cases, serious games can be an effective learning tool for agricultural education.

5.3.3.1. Statistical Analysis. The T-statistic and p values for the pretest and posttest improvements in each group were calculated using the paired sample t-test formula.

Control group:

$$\text{Mean difference} : 3.96 - 3.74 = 0.22,$$

$$\text{Standard deviation (SD)}: \pm 0.155,$$

$$\text{Standard error (SE)}: 0.155/\sqrt{20} \approx 0.0347,$$

$$T\text{-statistic} : 0.22/0.0347 \approx 6.35,$$

$$p \text{ value} : 0.0000043 \text{ (very significant, } p < 0.05).$$

Experimental group:

$$\text{Mean difference} : 4.56 - 4.40 = 0.16,$$

$$\text{Standard deviation (SD)}: \pm 0.113,$$

$$\text{Standard error (SE)}: 0.113/\sqrt{20} \approx 0.0253,$$

$$T\text{-statistic} : 0.16/0.0253 \approx 6.33,$$

$$p \text{ value} : 0.0000044 \text{ (very significant, } p < 0.05).$$

These values confirm that both groups showed statistically significant improvements; the improvements were not due to random chance. The serious game intervention is therefore considered adequate based on rigorous statistical testing.

5.3.3.2. t-Test Analysis. The statistical analysis presents the variance of both groups (control and experimental serious game) at the pretest and posttest stages. The descriptive results show that 20 students participated in each group. The mean and SD were calculated for each group. The control group had a pretest mean of 3.74 and a posttest mean of 3.96, while the experimental serious game group had a pretest mean of 4.4 and a posttest mean of 4.56. Similarly, the Tukey post hoc test showed significant pairwise differences between the experimental serious game group and the control group in the posttest. However, there was no significant difference in pretest scores between groups. The pairwise difference between the experimental serious game group and the control group in the posttest was 0.6 ($p < 0.05$), indicating that the serious game intervention had a statistically significant impact on learning outcomes.

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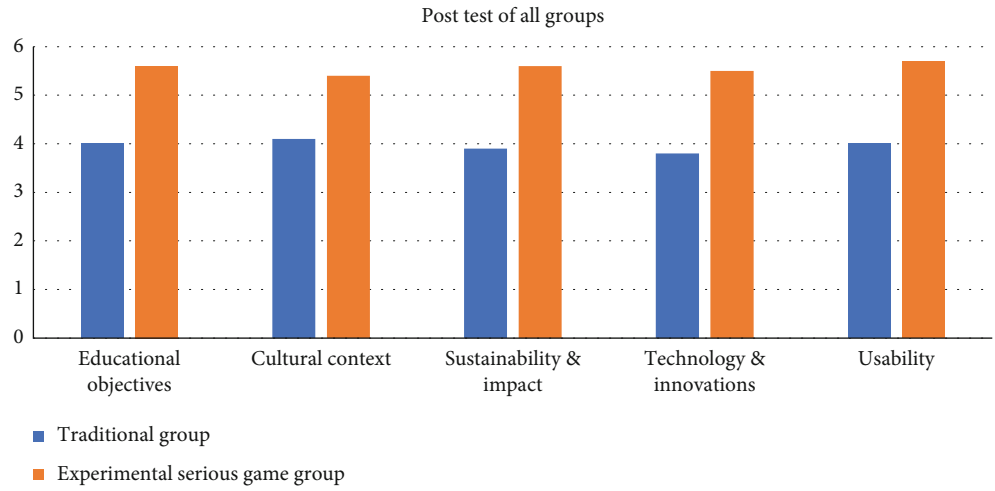


FIGURE 8: Bar chart of all posttest groups.

TABLE 5: Descriptive analysis of the control group.

Agrielements	Pretest mean	Posttest mean	Mean difference	Standard deviation (SD)
Educational objectives	3.8	4.0	+0.2	±0.141
Cultural context	3.9	4.1	+0.2	±0.141
Sustainability and impact	3.6	3.9	+0.3	±0.212
Technology and innovations	3.6	3.8	+0.2	±0.141
Usability	3.8	4.0	+0.2	±0.141
Overall mean	3.74	3.96	+0.22	±0.155

TABLE 6: Descriptive analysis of the experimental serious game group.

Agrielements	Pretest mean	Posttest mean	Mean difference	Standard deviation (SD)
Educational objectives	4.4	4.6	+0.2	±0.141
Cultural context	4.2	4.4	+0.2	±0.141
Sustainability and impact	4.5	4.6	+0.1	±0.070
Technology and innovations	4.3	4.5	+0.2	±0.141
Usability	4.6	4.7	+0.1	±0.070
Overall mean	4.4	4.56	+0.16	±0.113

TABLE 7: Comparison of both groups.

Metric	Control group	Experimental serious game group
Mean pretest score	3.74	4.4
Mean posttest score	3.96	4.56
Mean improvement	+0.22	+0.16
T-statistic	11.00	6.53
p value	0.000388	0.002838

6. Conclusion

This study examined the effectiveness of a culturally tailored game design model for smart agriculture, utilizing the SAG as a learning tool. The research was aimed at assessing how a serious game-based approach enhances knowledge and awareness of smart agriculture practices compared to traditional learning methods. A quasiexperimental design was employed, comparing a traditional learning group with an experimental serious game group. The pretest results showed no significant differences between the groups, confirming that participants had a similar baseline knowledge of smart agriculture. However, the posttest results substantially improved the experimental serious game group, particularly in precision farming, IoT-based solutions, and sustainable agricultural practices. t-test analysis confirmed that the SAG effectively enhanced participants' understanding compared to traditional methods. These findings emphasize the potential of culturally tailored serious games in advancing agricultural education. By integrating interactive and immersive elements, the SAG provided an engaging learning experience that helped participants better understand and retain complex

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agrarian concepts. The results suggest that serious games can be a valuable educational tool for training farmers, students, and agricultural professionals in adopting modern farming techniques. Future research can explore further enhancements in game design, the integration of real-world farming scenarios, and broader applications in diverse cultural and agricultural settings. This study underscores the importance of localized, game-based learning solutions in promoting sustainable and technology-driven agrarian practices.

7. Limitations and Future Work

While this research provides valuable insights into the effectiveness of a culturally tailored game design model for smart agriculture, several limitations should be acknowledged. The study was conducted with a specific group of participants, limiting its generalizability to wider farming communities or different cultural and educational settings. Future studies should include a larger and more diverse participant pool to validate the findings further. Additionally, the SAG primarily focused on precision farming, IoT applications, and sustainable practices, while overlooking other critical agricultural domains, such as livestock management, soil health, and climate adaptation techniques. Expanding the game content could enhance its effectiveness for a broader audience. Another limitation is the short-term evaluation, as the impact of the serious game was assessed immediately after gameplay. A longitudinal study would be necessary to determine whether knowledge retention and behavioral changes persist over time. Furthermore, the study assumed participants had adequate access to digital devices and a basic level of digital literacy. However, in many rural farming communities, limited access to technology and low digital literacy rates may hinder the game's scalability and impact. Lastly, the study only compared traditional learning with the SAG approach. Further research could compare SAG with other modern digital learning tools, such as mobile apps and virtual simulations, to assess its relative effectiveness. To enhance future experimental designs and minimize the need for manual pre- and postsurveys, we plan to integrate personalized in-game assessment mechanisms. These features will automatically monitor and evaluate student performance, learning progress, and concept mastery throughout gameplay. By embedding real-time analytics and adaptive feedback directly into the game, this form of personalization will provide a more tailored learning experience for each student while simultaneously enabling researchers to collect accurate and scalable data without the need for external surveys. Despite these limitations, the findings suggest a strong potential for serious games in advancing smart agriculture education. Future research should address these constraints and further explore the role of personalization in enhancing the applicability, scalability, and educational impact of game-based learning models in the agricultural sector.

Data Availability Statement

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

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